Fiscal Year:	FY 2009	**Task Last Updated:**	FY 05/12/2010
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Project Title:	Vigilance, Stress and Sleep/Wake Measures in NEEMO 14-A Simulated Space Environment		
Division Name:	Human Research		
Program/Discipline:	NSBRI		
Program/Discipline--Element/Subdiscipline:	NSBRI--Neurobehavioral and Psychosocial Factors Team		
Joint Agency Name:		**TechPort:**	No
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Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	Directed Research
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No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
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COI Name (Institution):			
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	NASA Extreme Environment Mission Operations (NEEMO) has astronauts live in the NOAA Aquarius habitat (19 meters underwater off the coast of Florida). Aquarius is an excellent space analog environment for a number of challenges in the area of behavior and performance: confinement, limited personal space, isolation, workload timeline demands, EVAs, environmental risk, no quick return to surface in an emergency. Aquarius is operated by the National Underwater Research Center of the UNC at Wilmington. The 9-day NEEMO 14 mission was co-sponsored by NASA and the National Space Biomedical Research Institute (NSBRI). The primary operational objectives of NEEMO 14 focused on advanced extravehicular activity (EVA) exploration, monitoring of behavioral health and performance in the crew, operational habitability and human factors, immune function assessment of the crew, effects of low autonomy on the crew and testing the usefulness of a new advanced scheduling and planning tool. This specific project concerned monitoring the behavioral health and		

Task Description:	performance of the NEEMO 14 crew during the mission, using the PVT Self Test. The NEEMO 14 Mission was originally scheduled for July-August of 2008, but then postponed by NASA to February-March 2009. It was subsequently postponed a second time by NASA. These postponements resulted in NASA BHP requesting we move our NEEMO 14 measures and goals to another space analog environment - the Haughton Mars Project (HMP) - that was scheduled to be conducted in the time frame of this project. In addition, we used astronauts who participated in our earlier studies in NEEMO 9, 12 and 13 to develop the PVT Self Test, to evaluate the prototypical PVT Self Test graphical, user-feedback interface of a new laptop computer-software based version of the 3-minute PVT Self Test. Consequently, the findings from this project involve the results from two projects that we completed at NASA's request to meet the goals we were planning on achieving in NEEMO 14 (which had been cancelled by NASA). Project 1. PVT Self Test data acquisition on N=9 participants during the HMP mission was successful. A total of 112 PVT performance tests were acquired and sleep-wake diary and subjective fatigue and stress scales were successfully acquired electronically during the morning and evening PVT Self Test sessions. Actigraph data was successfully collected for 12-31 days per participant, (depending on how long the participant was present at HMP). Facial video data for optical computer recognition (OCR) of stress was successfully collected on 3 PVT Self Test sessions, but it was determined from thee data that it was not feasible in the HMP environment to collect this data via a small portable camera, as there was rarely space or time to set the camera up to record the participant during the PVT Self Test sessions. This has led us to develop the ability to automatically record video of the face from the built in computer webcam. Project 2. N=6 past NEEMO astronauts who completed >200 PVT Self Test trials during NEEMO 9, 12 and 13 missions agreed to evaluate the PVT Self Test. Their comments regarding screen layout, keyboard functionality and wording and layout of questionnaires were integrated into the final development of the PVT Self Test. All of the N=6 evaluating astronauts were enthused about the development of a brief, short-duration performance test that could be used to get immediate objective feedback of their current performance levels on the test. This is the final report for this project, which accomplished three key goals: (1) critical astronaut input was obtained on the features to develop for the PVT Self Test performance feedback, for its use in space (on the International Space Station); (2) evidence that the PVT Self test was sensitive to the effects of psychoactive medications; and (3) evidence that a computer-based optical data acquisition system was needed to acquire video for the continued testing of optical computer recognitions algorithms of stress and negative states in operational environments.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Development of a 3-minute PVT performance test that is a very sensitive objective measure of fatigue and sleepiness in astronauts during space flight, will also have extensive utility in a wide range of activities on Earth, including evaluating readiness to perform in military personnel, transportation and security personnel and people working in high risk operational environments (e.g., power plants), as well as use in identifying excessive sleepiness in patients.
Task Progress:	The NEEMO 14 mission to be used for this project was originally scheduled for July-August of 2008, but then postponed by NASA to February-March 2009. It was subsequently postponed a second time by NASA. These postponements resulted in the project measures and goals being implemented in another space analog environment - the Haughton Mars Project (HMP) on Devon Island, Canada. At HMP, PVT Self Test performance data acquisition was completed on N=9 participants during the HMP mission. A total of 112 PVT performance tests were acquired and sleep-wake diary and subjective fatigue and stress scales were successfully acquired electronically during the morning and evening PVT Self Test sessions. Actigraph data was successfully collected for 12-31 days per participant, (depending on how long the participant was present at HMP). Facial video data for optical computer recognition (OCR) of stress was successfully collected on 3 PVT Self Test sessions, but it was determined from these data that it was not feasible in the HMP environment to collect this data via a small portable camera, as there was rarely space or time to set the camera up to record the participant during the PVT Self Test sessions. This has led us to develop the ability to automatically record video of the face from the built in computer webcam. In addition to data acquisition at HMP, we were given permission to contact astronauts who participated in our earlier studies in NEEMO 9, 12 and 13 providing PVT Self Test data, to evaluate the prototypical PVT Self Test graphical, feedback interface of a new laptop computer-software-based version of the 3-minute PVT Self Test. A total of 6 NEEMO participants who had completed >200 PVT Self Test trials during NEEMO 9, 12 and 13 missions agreed to evaluate the PVT Self Test feedback interface. Their comments regarding screen layout, keyboard functionality and wording and layout of questionnaires were integrated into the final development of the PVT Self Test. All 6 of the astronauts were enthusiastic about the development of a brief, short-duration performance test that could be used to get immediate objective feedback of their current performance levels on the test.
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